Meng Zhao

Postdoctoral Researcher | CY Cergy Paris University

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RESEARCH INTERESTS

Stochastic PDEs and fluid dynamics; ergodicity and mixing; regularization by noise; large deviations.

ACADEMIC APPOINTMENTS

Postdoctoral Researcher 2026–present
AGM (Analyse, géométrie et modélisation), CY Cergy Paris University, France.
Research mentor: Prof. Armen Shirikyan.

Visiting Ph.D. Researcher 2024
Institut de Mathématiques de Jussieu–Paris Rive Gauche, Université Paris Cité, France.
Host: Prof. Sergei Kuksin.
AGM, CY Cergy Paris University, France.
Host: Prof. Armen Shirikyan.

EDUCATION

Ph.D. in Mathematics December 2025
Shanghai Jiao Tong University, Shanghai, China (2021–2025).
Advisors: Prof. Ya-Guang Wang and Prof. Vahagn Nersesyan (joint supervision).

M.Sc. in Mathematics 2021
Shanghai Jiao Tong University, Shanghai, China. Advisor: Prof. Ya-Guang Wang.

M.Sc. in Finance 2018
Tongji University, Shanghai, China.

B.Sc. in Economics 2016
Hunan University, Changsha, China.

PREPRINTS

- **Meng Zhao**, *Polynomial mixing for infinite-energy solutions of the white-forced Navier–Stokes equation in an infinite pipe*, arXiv:2610.15216 (2026).
- **Meng Zhao**, *Delayed blow-up by transport noise for the 3D Navier–Stokes equation with Navier-slip boundary conditions*, arXiv:2606.19060 (2026).
- **Meng Zhao**, *Level-3 large deviations for the white-forced 2D Navier–Stokes system in a bounded domain*, arXiv:2506.14119 (2025).
- Vahagn Nersesyan and **Meng Zhao**, *Polynomial mixing for the white-forced Navier–Stokes system in the whole space*, arXiv:2410.15727 (2024).
- Ya-Guang Wang and **Meng Zhao**, *Local and global existence for the stochastic Prandtl equation driven by multiplicative noises in two and three dimensions*, arXiv:2408.04546 (2024).

JOURNAL ARTICLES

- Vahagn Nersesyan and **Meng Zhao**, *Exponential mixing for the white-forced complex Ginzburg–Landau equation in the whole space*, *SIAM J. Math. Anal.* **56** (2024), no. 3, 3646–3678.
- Ya-Guang Wang and **Meng Zhao**, *On Kato's conditions for the inviscid limit of the two-dimensional stochastic Navier–Stokes equation*, *J. Math. Phys.* **65** (2024), no. 8, 081507, 16 pp.

SELECTED TALKS

- *Ergodicity and mixing properties for white-forced dissipative PDEs in unbounded domains*, 14th AIMS Conference, NYU Abu Dhabi, United Arab Emirates, December 16, 2024.
- *Well-posedness and regularization by noise for the stochastic Prandtl equation*, 19th International Conference on Hyperbolic Problems: Theory, Numerics and Applications, Shanghai Jiao Tong University, July 2, 2024.
- *On Kato's conditions for the inviscid limit of the two-dimensional stochastic Navier–Stokes equation*, 10th International Congress on Industrial and Applied Mathematics, Waseda University, Tokyo, August 23, 2023.
- *On Kato's conditions for the inviscid limit of the two-dimensional stochastic Navier–Stokes equation*, International Conference on Hyperbolic Partial Differential Equations, Shanghai Jiao Tong University, August 4, 2023.

AWARDS

National Scholarship for Postgraduate Students. 2024

Second Place, Student Paper Competition, 14th AIMS Conference. 2024

Wu Wen-Tsun Scholarship for Outstanding Doctoral Students. 2021

National Scholarship for Postgraduate Students. 2020

TEACHING

Teaching Assistant 2019–2025
Shanghai Jiao Tong University; assisted courses taught by Profs. James Fullwood and Xin Chen.